# B3.1 Gas exchange

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 52 |

**About 50 minutes,** or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make. Look at the bold command term: "state" needs a brief answer, while "explain" needs reasons.

## Section A: Recall

9 MARKS

1 **LIST** four properties of gas-exchange surfaces. [B3.1.2] [2]

2 **STATE** the effect of surfactant on the lining of an alveolus. [B3.1.4] [1]

3 **STATE** the structures in the lungs where gas exchange takes place. [B3.1.4] [1]

4 **DISTINGUISH** between tidal volume and vital capacity. [B3.1.6] [2]

5 **STATE** the two sets of muscles that contract during inhalation at rest. [B3.1.5] [2]

6 **STATE** the equation that gives vital capacity from the other measured volumes. [B3.1.6] [1]

## Section B: Understanding

14 MARKS

7 **OUTLINE** why gas exchange becomes more challenging as organisms increase in size. [B3.1.1] [2]

8 **EXPLAIN** three ways in which concentration gradients are maintained at the surface of the lungs. [B3.1.3] [3]

9 **EXPLAIN** how the alveoli and their blood supply are adapted for gas exchange. [B3.1.4] [4]

10 **EXPLAIN** why a fish must keep water moving over its gills. [B3.1.3] [2]

11 A student wrote: "Air rushes into the lungs, and this pushes the diaphragm down." **EXPLAIN** the mistake and give the correct sequence. [B3.1.5] [3]

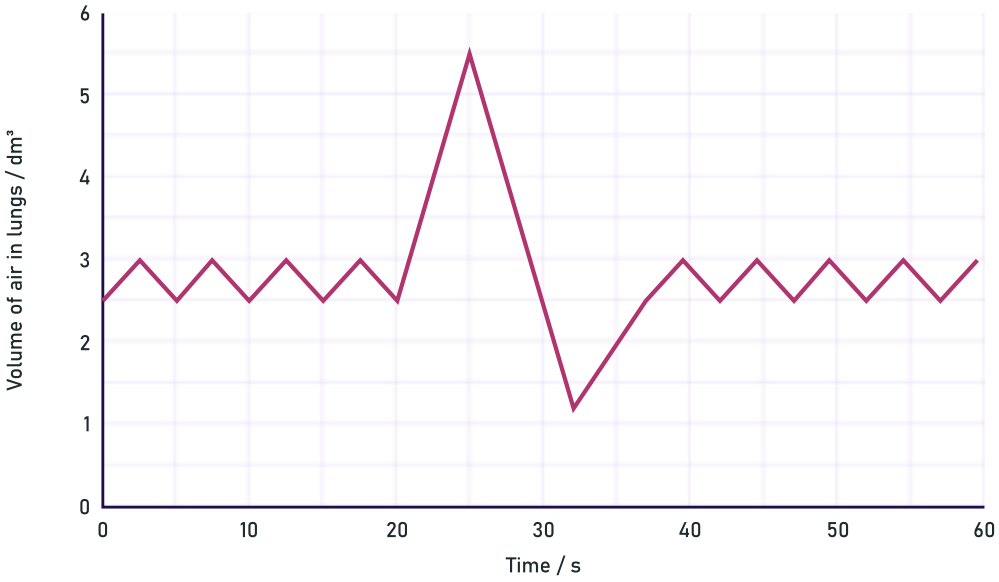
## Section C: Working with data

21 MARKS

*Data in questions 12 and 14 are invented for practice. The patterns are realistic, but they do not describe a real person.*

**12** [B3.1.6]

The spirometer trace shows a person breathing normally, then taking the deepest possible breath in, then breathing out as far as possible, then breathing normally again.




**a** DETERMINE the tidal volume. [1]

**b** DETERMINE the inspiratory reserve. [1]

**c** DETERMINE the expiratory reserve. [1]

**d** CALCULATE the vital capacity, showing your working. [1]

**e** CALCULATE the resting ventilation rate in breaths per minute. [2]

**f** PREDICT two changes to the trace during vigorous exercise. [2]

**13** [B3.1.3, B3.1.4]

The table gives typical percentages of two gases in air at three points.

| GAS | INHALED AIR / % | AIR IN ALVEOLI / % | EXHALED AIR / % |
|---|---|---|---|
| OXYGEN | 20.9 | 14.0 | 16.0 |
| CARBON DIOXIDE | 0.04 | 5.5 | 4.0 |

a CALCULATE the percentage change in oxygen from inhaled air to air in the alveoli. [2]

b EXPLAIN why the carbon dioxide percentage is far higher in the alveoli than in inhaled air. [2]

c SUGGEST why exhaled air contains more oxygen than air in the alveoli. [2]

**14** [B3.1.5, B3.1.6]

A student measured their ventilation at three levels of activity.

| ACTIVITY | VENTILATION RATE / BREATHS PER MINUTE | TIDAL VOLUME / $DM^3$ | AIR BREATHED / $DM^3$ PER MINUTE |
|---|---|---|---|
| REST | 12 | 0.5 | |
| WALKING | 20 | 1.0 | |
| RUNNING | 38 | 2.0 | |

a CALCULATE the three values missing from the table. [2]

b CALCULATE the percentage change in air breathed per minute from rest to running. [2]

c EXPLAIN why air breathed per minute rises during exercise. [3]

## Section D: Longer answers

8 MARKS

**15** DESCRIBE how the movements of muscles and ribs cause air to enter and then leave the lungs. [B3.1.5] [6]

16 **EVALUATE** using a balloon, rather than a spirometer, to estimate vital capacity. [B3.1.6] [2]